

(1001CJA102120024)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

19-07-2020

**JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)****ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

| SECTION-I  | Q. | 1   | 2       | 3   | 4  | 5  | 6   | 7   | 8     | 9   | 10  |
|------------|----|-----|---------|-----|----|----|-----|-----|-------|-----|-----|
|            | A. | A   | B       | C   | A  | D  | A,D | A,C | B,C,D | A,D | B,C |
| SECTION-II | Q. | 11  | 12      | 13  | 14 | 15 | 16  | 17  | 18    | 19  | 20  |
|            | A. | C,D | A,B,C,D | A,C |    |    |     |     |       |     |     |

**PART-2 : CHEMISTRY**

| SECTION-I  | Q. | 1   | 2   | 3   | 4  | 5  | 6       | 7   | 8     | 9       | 10    |
|------------|----|-----|-----|-----|----|----|---------|-----|-------|---------|-------|
|            | A. | A   | B   | D   | B  | B  | A,B,C,D | B,D | A,B,D | A,B,C,D | A,B,C |
| SECTION-II | Q. | 11  | 12  | 13  | 14 | 15 | 16      | 17  | 18    | 19      | 20    |
|            | A. | A,D | A,D | A,B |    |    |         |     |       |         |       |

**PART-3 : MATHEMATICS**

| SECTION-I  | Q. | 1   | 2     | 3   | 4  | 5  | 6   | 7     | 8   | 9   | 10    |
|------------|----|-----|-------|-----|----|----|-----|-------|-----|-----|-------|
|            | A. | A   | B     | D   | C  | B  | A,B | A,B,D | A,B | A,B | A,B,C |
| SECTION-II | Q. | 11  | 12    | 13  | 14 | 15 | 16  | 17    | 18  | 19  | 20    |
|            | A. | A,C | A,B,D | A,B |    |    |     |       |     |     |       |

(1001CJA102120025)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

REVIEW TEST

19-07-2020

**JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)****ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

| SECTION-I  | Q. | 1   | 2   | 3   | 4   | 5  | 6  | 7   | 8   | 9     | 10    |
|------------|----|-----|-----|-----|-----|----|----|-----|-----|-------|-------|
|            | A. | B   | C   | D   | C   | C  | C  | A,C | B,D | A,B,D | A,B,D |
| SECTION-II | Q. | 11  | 12  | 13  | 14  | 15 | 16 | 17  | 18  | 19    | 20    |
|            | A. | A,C | B,C | A,C | A,D | B  | C  | A   | B   |       |       |

**PART-2 : CHEMISTRY**

| SECTION-I  | Q. | 1   | 2     | 3   | 4  | 5  | 6  | 7     | 8   | 9     | 10    |
|------------|----|-----|-------|-----|----|----|----|-------|-----|-------|-------|
|            | A. | C   | B     | B   | B  | B  | A  | A,B,C | A,D | A,B,D | A,C,D |
| SECTION-II | Q. | 11  | 12    | 13  | 14 | 15 | 16 | 17    | 18  | 19    | 20    |
|            | A. | B,C | A,B,C | A,B | A  | B  | C  | B     | A   |       |       |

**PART-3 : MATHEMATICS**

| SECTION-I  | Q. | 1     | 2     | 3     | 4     | 5  | 6  | 7       | 8       | 9   | 10  |
|------------|----|-------|-------|-------|-------|----|----|---------|---------|-----|-----|
|            | A. | A     | D     | D     | B     | D  | A  | A,B,C,D | A,B,C,D | A,D | A,C |
| SECTION-II | Q. | 11    | 12    | 13    | 14    | 15 | 16 | 17      | 18      | 19  | 20  |
|            | A. | A,B,C | A,B,D | A,B,C | A,C,D | A  | C  | B       | C       |     |     |

**JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)****PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (A)****Sol.** For flux linked with A, we use

$$\phi_A = N_1 \phi_1 = L_A i_1$$

$$\Rightarrow L_A = \frac{N_1 \phi_1}{i_1} = \frac{12500 \times 0.6 \times 10^{-3}}{5} = 1.5 \text{ H}$$

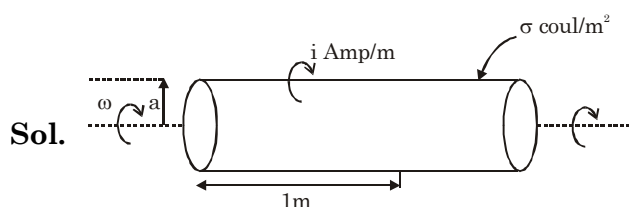
Similarly we use for coil B  $\phi_B = N_2 \phi_2 = L_B i_2$ 

$$\Rightarrow L_B = \frac{N_2 \phi_2}{i_2} = \frac{16000 \times 0.8 \times 10^{-3}}{5} = 2.56 \text{ H}$$

If M is the mutual ind coeffi between coils, we use  $\phi_B = M i_A$ 

$$\Rightarrow M = \frac{\phi_B}{i_A} = \frac{N_2 (0.6 \phi_A)}{i_A}$$

$$= \frac{16000 \times 0.6 \times 0.6 \times 10^{-3}}{5} = 1.152 \text{ H}$$

**2. Ans. (B)****Sol.**In rotating cylinder,  
Current per unit length of cylinder is  $i = \sigma$ .

$$(1.2 \pi a) \cdot \frac{\omega}{2\pi} = \sigma a \omega$$

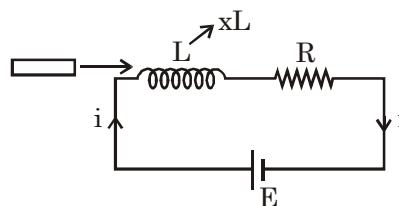
Magnetic field inside the cylinder due to circulating current in  $B = \mu_0 i = \mu_0 \sigma a \omega$ 

Magnetic energy density

$$u_m = \frac{B^2}{2\mu_0} = \frac{1}{2} \mu_0 \sigma^2 a^2 \omega^2$$

Total energy stored per unit length of cylinder (inside) is  $U = U_m \cdot (1. \pi a^2)$ 

$$U = \frac{1}{2} \pi \mu_0 \sigma^2 a^4 \omega^2$$

**3. Ans. (C)****Sol.** At  $t = 0$ , Initial current  $I_0 = \frac{E}{R}$ flux through coil  $\phi = L I_0$ at  $t = 0 +$ , just after insertion of rod if circuit current charges to  $I_1$ 

$$\Rightarrow \phi_1 = \phi_f$$

$$L I_0 = n L I_1$$

$$I_1 = \frac{I_0}{n}$$

After  $t = 0$  if  $i$  is circuit current then using KVL.

$$E = -nL \frac{di}{dt} - iR = 0$$

$$\int_{E/nR}^i \frac{Li}{E - iR} = \int_0^t \frac{dt}{nL}$$

$$\frac{1}{R} \ln \left( \frac{E - iR}{E \left( 1 - \frac{1}{n} \right)} \right) = \frac{t}{nL}$$

$$i = \frac{E}{R} \left[ 1 - \left( 1 - \frac{1}{n} \right) e^{-\frac{Rt}{nL}} \right]$$

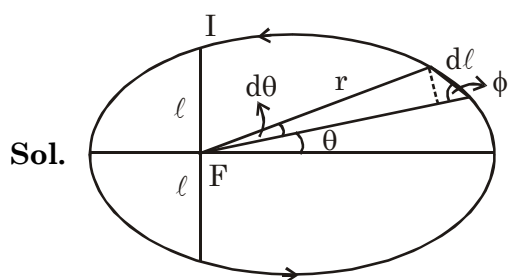
**4. Ans. (A)****Sol.** Charge flown through coil

$$\Delta q = \frac{|\Delta \phi|}{R}$$

Initial flux the coil  $\phi_i = B \pi a^2$   
Final flux the coil  $\phi_f = -B \pi a^2$  ]  $|\Delta \phi| = 2B \pi a^2$ 

$$\text{Charge flown } \Delta q = \frac{2B \pi a^2}{R}$$

5. Ans. (D)



$$\frac{\ell}{r} = (1 + e \cos \theta) \quad \dots(i)$$

$$dB = \left( \frac{\mu_0}{4\pi} \right) \frac{Id\vec{\ell} \times \vec{r}}{r^3}$$

$$dB = \frac{\mu_0}{4\pi} \frac{Idr \sin \phi}{r^3}$$

$$d\ell \sin \phi = r d\theta$$

$$dB = \frac{\mu_0}{4\pi} \frac{I}{r^2} r d\theta = \frac{\mu_0}{4\pi} \frac{I}{r} d\theta$$

From equation (i)

$$dB = \frac{\mu_0}{4\pi} I \left\{ \frac{1 + e \cos \theta}{\ell} \right\} d\theta$$

$$\Rightarrow B = \frac{\mu_0 I}{4\pi \ell} \int_0^{2\pi} (1 + e \cos \theta) d\theta$$

$$B = \frac{\mu_0}{4\pi \ell} I [\theta + e \sin \theta]_0^{2\pi} \Rightarrow \frac{\mu_0}{4\pi \ell} I \times 2\pi \Rightarrow \frac{\mu_0}{2} \frac{I}{\ell}$$

6. Ans. (A, D)

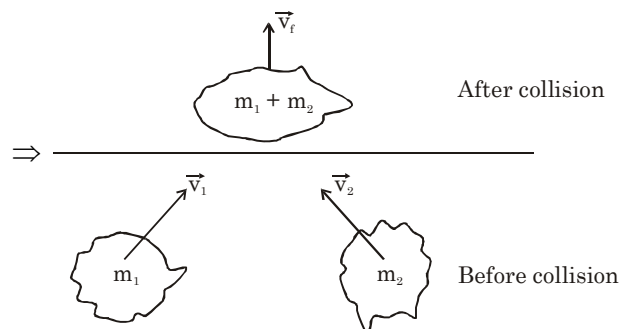
Sol. Concept:

- Since the two bodies stick to each other after the collision so the velocity of center of mass of the system will be equal to the final velocity.
- Second part of the question is more interesting than the first. In the second part we wish to find out the unit vector along the line of impact. For this, let's first understand the formal definition of "line of impact". Line of impact during the collision is the line in space along which the two bodies exchange the "Impulse" with each other.

By Newton's Third Law we also know that the impulses exchanged by the two bodies will be equal and opposite to each other, so if we calculate the net impulse on one body then it will be equal and opposite to the impulse experienced by the second body.

So, calculating net impulse on just one body is sufficient to solve the question.

- By Impulse Momentum theorem we know that Impulse Experienced by a body is equal to the net change in momentum of the body. We will apply this idea on any of the bodies to calculate the net change in momentum and that will be the direction of net impulse on that body.



$$\Rightarrow (a) m_1 \vec{v}_1 + m_2 \vec{v}_2 = (m_1 + m_2) \vec{v}_f$$

$$\Rightarrow \vec{v}_f = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$$

$$(b) \text{ Impulse on body (i) } = \vec{I}_1 \text{ (say)}$$

$$\Rightarrow \vec{I}_1 = m_1 \vec{v}_f - m_1 \vec{v}_1$$

$$\text{and impulse on body (ii) } = \vec{I}_2 \text{ (say)}$$

$$\Rightarrow \vec{I}_2 = m_2 \vec{v}_f - m_2 \vec{v}_2$$

$$\Rightarrow \vec{I}_1 = \frac{m_1 m_2 \vec{v}_1 + m_1 m_2 \vec{v}_2}{m_1 + m_2} - m_1 \vec{v}_1$$

$$= \frac{m_1 m_2}{m_1 + m_2} (\vec{v}_2 - \vec{v}_1) = -\vec{I}_2$$

(By calculation we can see)

$$\Rightarrow \text{Now answer to part (b)}$$

$$\Rightarrow \frac{\vec{I}_1}{|\vec{I}_1|} \text{ or } \frac{\vec{I}_2}{|\vec{I}_2|}$$

7. Ans. (A,C)

Sol. Since the ring is superconducting so it will not allow the change of flux through it. In a "perfect conductor" there is no resistance whatever to the current. So if currents are generated in it, they can keep going forever. In fact, the *slightest* emf would generate an arbitrarily large current-which really means that there can be no emf's at all. Any attempt to make a magnetic flux

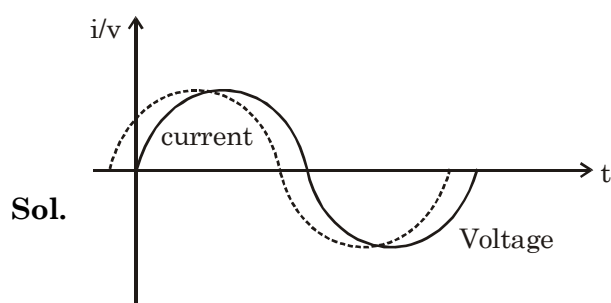
go through such a sheet generates currents that create opposite B fields-all with infinitesimal emfs, so with no flux entering. Now, the flux due to external field and flux due to induced currents will be equal and opposite through the ring, so they will cancel each other. Since the two fluxes will be equal in magnitude so the flux due to induced currents through the ring can be calculated and that will be equal to:

$\phi = LI_{\text{Induced}}$ , Which will lead to the following equation.

$$\phi = Kt\pi R^2 = LI_{\text{Induced}}$$

And, here we have the induced current.

8. **Ans. (B, C, D)**



Sol.

The graph shows V & I as function of time

Current leads the voltage by  $\frac{\pi}{3}$

Power is positive if V & I are of same sign.  
Power is negative if V & I are of opposite sign.

If  $\omega \uparrow \Rightarrow \frac{1}{\omega C} \downarrow$  thus angle decreases.

9. **Ans. (A,D)**

Sol. Superposition of waves (i) & (iii) will give travelling wave having amplitude of  $a\sqrt{2}$  (waves are along x-axis but particle displacement are along y & z-axis respectively)

$$z_1 + z_2 = a \left[ \sin \omega \left( t - \frac{x}{v} \right) + \sin \left\{ \omega \left( t + \frac{x}{v} \right) + \frac{\pi}{2} \right\} \right]$$

10. **Ans. (B,C)**

Sol.  $\therefore \lambda = 3m$   $\therefore k = \frac{2\pi}{\lambda} = \frac{2\pi}{3}$

Maximum displacement in half time period  
=  $2a = 10$  m

So maximum average velocity

$$= \frac{10}{\frac{T}{2}} = 5 \Rightarrow T = 4s \Rightarrow \omega = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$$

11. **Ans. (C,D)**

Sol. Equation of SHM of particle who is at antinode is  $y = 2A \sin \left( \frac{2\pi}{T} \right) t$  at time  $t = \frac{T}{8}$ .

$y = 2A \sin \frac{\pi}{4} = \sqrt{2} A$ ; Displacement of particle at note is always zero.

12. **Ans. (A,B,C,D)**

Sol.  $\vec{F} = q(\vec{v} \times \vec{B})$ ,  $\vec{v} = 10^5 \hat{i}$ ,  $\vec{B} = -2 \times 10^{-3} \hat{k}$

$\Rightarrow$  Force will be in y-direction

$\Rightarrow$  Motion of particle will be in xy plane

Time period

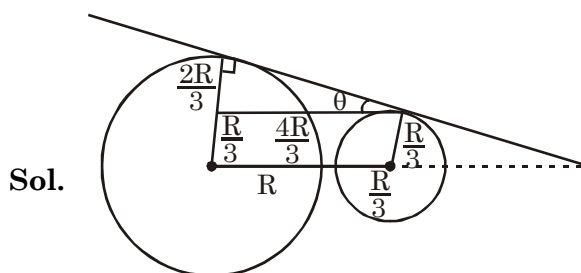
$$T = \frac{2\pi m}{qB} = \frac{2\pi}{\pi \times 10^{10} \times 2 \times 10^{-3}} = 10^{-7} \text{ s}$$

Radius of path

$$\frac{mv}{qB} = \frac{10^5}{\pi \times 10^{10} \times 2 \times 10^{-3}} = \frac{5}{\pi} \times 10^{-3} \text{ m} = \frac{5}{\pi} \text{ mm}$$

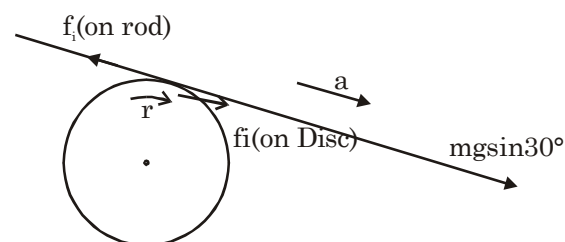
At  $t = \frac{T}{4} = \frac{10^{-7}}{4} \text{ s}$ . Velocity of particle will be in +y direction

13. **Ans. (A, C)**



Sol.

$$\sin \theta = \frac{2R/3}{4R/3} = \frac{1}{2} \Rightarrow \theta = 30^\circ$$



$$mg \sin 30 - (f_1 + f_2 + f_3 + \dots \infty) = ma \dots (1)$$

for  $i^{\text{th}}$  Disc

$$\tau_i = I_i \alpha \rightarrow f_i r_i = \left( \frac{m_i r_i^2}{2} \right) \left( \frac{a}{r_i} \right)$$

$$\therefore f_i = \frac{m_i a}{2} \quad \dots(2)$$

From equation (1) & (2)

$$mg \sin 30 - \frac{a}{2} \left( \sum m_i \right) = ma \quad \dots(3)$$

since  $m_{\text{disc}} \propto \text{Area}$  so if  $m_1 = m$ ,  $m_2 = \frac{m}{9}$ ,

$$m_3 = \frac{m}{81}$$

From eq (3)

$$\frac{mg}{2} - \frac{a}{2} \left\{ m + \frac{m}{9} + \frac{m}{81} + \dots \right\} = ma$$

$$\frac{mg}{2} = ma + \frac{a}{2} \left\{ \frac{m}{1 - \frac{1}{9}} \right\} = ma + \frac{ma}{2} \times \frac{9}{8}$$

$$= \frac{25}{16} ma$$

## SECTION-II

1. Ans. 1.00

Sol. The motional emf in rod PQ is  $e_{PQ} = Bv\ell$

charge on cap is  $q = C(Bv\ell) \Rightarrow i = \frac{dq}{dt} = CB\ell a$

If  $a$  is acceleration of rod PQ, we use

$$kx - Bi\ell = ma$$

$$kx = ma + B^2\ell^2 Ca$$

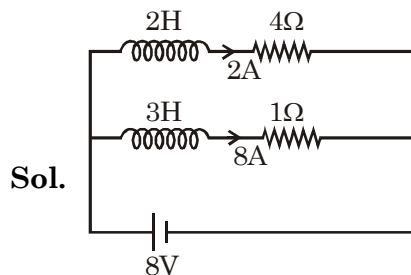
$$a = - \left( \frac{k}{m + CB^2\ell^2} \right) x$$

Comparing it with  $a = -\omega^2 x$

$$\text{We get } \omega = \sqrt{\frac{k}{m + B^2\ell^2 C}}$$

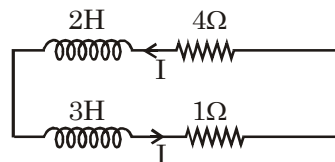
$$\Rightarrow T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m + CB^2\ell^2}{k}}$$

2. Ans. 4.00



Sol.

At steady state (initially)



Just after opening the switch.

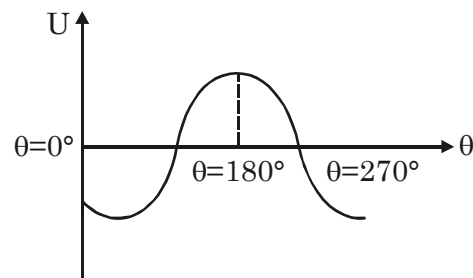
Applying flux conservation in the loop containing inductors and resistances.

$$\phi_i = \phi_f$$

$$3 \times 8 - 2 \times 2 = (3 + 2) I$$

$$I = \frac{24 - 4}{5} = \frac{20}{5} = 4 \text{ amp}$$

3. Ans. 9.00



Sol.

to reach  $\theta = 270^\circ$ , it has to cross the potential energy barrier at  $\theta = 180^\circ$  and to cross  $\theta = 180^\circ$  angular velocity at  $\theta = 180^\circ$  should be  $0^+$

$$k_i + U_i = k_f + U_f$$

$$\frac{1}{2} \left( \frac{3}{2} MR^2 \right) \omega^2 + (-MiAB \cos 0^\circ) = 0 + (-NiAB \cos 180^\circ)$$

$$\omega = \sqrt{80} \approx 9 \text{ rad/sec.}$$

4. Ans. 2.00

$$\text{Sol. } P_{\text{d.c.}} = \frac{V_0^2}{R}$$

$$P_{\text{a.c.}} = \frac{V_0^2}{\sqrt{R^2 + \omega^2 L^2}} \cdot \frac{R}{\sqrt{R^2 + \omega^2 L^2}} = \frac{v_0^2 / R}{1 + \left( \frac{\omega L}{R} \right)^2} = \frac{P_{\text{d.c.}}}{\eta}$$

$$\text{Thus } \frac{\omega L}{R} = \sqrt{\eta - 1}$$

$$\text{or } \omega = \frac{R}{L} \sqrt{\eta - 1}$$

$$v = \frac{R}{2\pi L} \sqrt{\eta - 1} = 2\text{kHz} \text{ of on putting the values.}$$

5. **Ans. 6.00**

$$\text{Sol. } f = n \left( \frac{v}{2\ell} \right)$$

$$\therefore \frac{n_1}{n_2} = \frac{\ell_1}{\ell_2} = \frac{3}{2}$$

$$\therefore \text{ minimum } n_1 = 3$$

$$\therefore f_{\min} = \frac{n_1}{2\ell_1} \sqrt{\frac{T}{\mu}}$$

$$= \frac{3}{2 \times 3} \sqrt{\frac{144}{1}}$$

$$= \frac{1}{2} \times 12$$

$$= 6$$

## PART-2 : CHEMISTRY

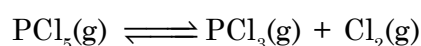
## SOLUTION

### SECTION-I

1. **Ans.(A)**
2. **Ans.(B)**
3. **Ans.(D)**
4. **Ans. (B)**
5. **Ans.(B)**
6. **Ans.(A,B,C,D)**
7. **Ans.(B,D)**
8. **Ans.(A,B,D)**
9. **Ans.(A,B,C,D)**
10. **Ans.(A,B,C)**
11. **Ans.(A,D)**
12. **Ans.(A,D)**
13. **Ans.(A,B)**

### SECTION-II

1. **Ans.(3.00)**
2. **Ans.(4.00)**
3. **Ans.(4.00)**
4. **Ans.(3.00)**



$$\begin{array}{ccc} 0.1 & - & - \\ \frac{0.1}{2} & \frac{0.1}{2} & \frac{0.1}{2} \end{array}$$

If  $P_{\text{PCl}_5} + P_{\text{PCl}_3} + P_{\text{Cl}_2} = P$ , then

$$K_p = 1.6 = \frac{\frac{0.1}{2} P \times \frac{0.1}{2} P}{\frac{0.1}{2} \times P \times \frac{0.3}{2}} = \frac{P}{3}$$

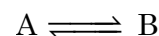
$$P = 4.8 \text{ bar}$$

$$P_{\text{solvent}} = 5.4 - 4.8 = 0.6 \text{ bar}$$

$$P \propto n \Rightarrow \frac{P}{P_{\text{solvent}}} = \frac{n}{n_{\text{solvent}}} \Rightarrow \frac{4.8}{0.6} = \frac{0.3/2}{n_{\text{solvent}}}$$

$$W_{\text{solvent}} = \frac{0.3}{16} \times 160 = 3\text{gm}$$

5. **Ans.(66.60 to 66.70)**



$$\begin{array}{ccc} t = 0 & 100 & 0 \\ t = 1 \text{ hr} & 80 & 20 \\ \text{At equi.} & 40 & 60 \end{array}$$

$$(K_1 + K_2) = \frac{1}{t} \ln \left( \frac{x_e}{x_e - x} \right)$$

$$\frac{1}{1} \ln \left( \frac{60}{60 - 20} \right) = \frac{1}{2} \ln \left( \frac{60}{60 - x} \right)$$

$$x = \frac{100}{3}$$

$$\% \text{ of A} = \frac{200}{3} = 66.66\%$$

**PART-3 : MATHEMATICS**

**SOLUTION**

**SECTION-I**

1. **Ans. (A)**

**Sol.** 
$$\vec{r} = \frac{(\vec{r} \cdot (\hat{n}_2 \times \hat{n}_3))\hat{n}_1 + (\vec{r} \cdot (\hat{n}_3 \times \hat{n}_1))\hat{n}_2 + (\vec{r} \cdot (\hat{n}_1 \times \hat{n}_2))\hat{n}_3}{[\hat{n}_1 \hat{n}_2 \hat{n}_3]}$$
$$= \frac{\hat{n}_1 + 2\hat{n}_2 + 3\hat{n}_3}{[\hat{n}_1 \hat{n}_2 \hat{n}_3]}$$

2. **Ans. (B)**

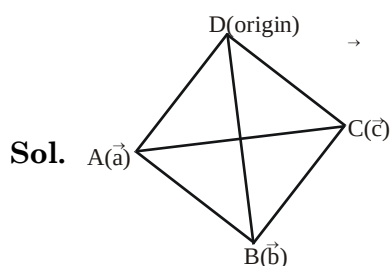
**Sol.** Let  $|\vec{a}| = |\vec{b}| = |\vec{c}| = |\vec{d}| = x$

Consider

$$(\vec{a} \times \vec{c}) \cdot (\vec{b} \times \vec{d}) = (\vec{c} \cdot \vec{d})(\vec{a} \cdot \vec{b}) - (\vec{c} \cdot \vec{b})(\vec{a} \cdot \vec{d}) = 0$$

$$\left\{ \because \vec{a} \cdot \vec{b} = \vec{c} \cdot \vec{d} = \vec{b} \cdot \vec{c} = \vec{a} \cdot \vec{d} = \frac{x^2}{2} \right\}$$

3. **Ans. (D)**



$$|\vec{a} - \vec{b}| = |-\vec{c}|$$

$$\Rightarrow \vec{a}^2 + \vec{b}^2 - \vec{c}^2 - 2\vec{a} \cdot \vec{b} = 0$$

$$\vec{b}^2 + \vec{c}^2 - \vec{a}^2 - 2\vec{b} \cdot \vec{c} = 0$$

$$\vec{c}^2 + \vec{a}^2 - \vec{b}^2 - 2\vec{a} \cdot \vec{c} = 0$$

$$\vec{a}^2 + \vec{b}^2 + \vec{c}^2 - 2\vec{a} \cdot \vec{b} - 2\vec{b} \cdot \vec{c} - 2\vec{c} \cdot \vec{a} = 0 \dots (1)$$

Also  $|\vec{a} + \vec{b} - \vec{c}|^2 \geq 0$

$$\vec{a}^2 + \vec{b}^2 + \vec{c}^2 + 2\vec{a} \cdot \vec{b} - 2\vec{a} \cdot \vec{c} - 2\vec{b} \cdot \vec{c} \geq 0 \dots (2)$$

(1) & (2) similarly

$$\vec{a} \cdot \vec{b} > 0$$

$$\vec{b} \cdot \vec{c} > 0$$

$$\vec{c} \cdot \vec{a} > 0$$

equality cannot hold as points becomes collinear.

4. **Ans. (C)**

**Sol.** 
$$\lim_{x \rightarrow \infty} x^2 \tan \left( \ln \left( \cos \frac{\pi}{x} \right)^{1/2} \right)$$

$$= \lim_{x \rightarrow \infty} x^2 \left( \ln \left( \cos \frac{\pi}{x} \right)^{1/2} \right)$$

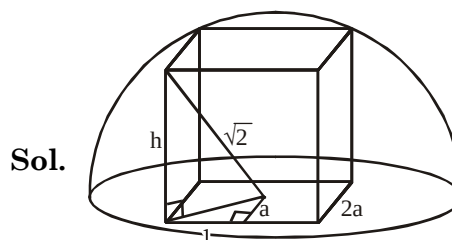
$$= \lim_{x \rightarrow \infty} \frac{x^2}{2} \ln \left( \cos \frac{\pi}{x} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{x^2}{2} \left( \ln \left( 1 + \left( \cos \frac{\pi}{x} - 1 \right) \right) \right)$$

$$= \lim_{x \rightarrow \infty} \frac{x^2}{2} \left( \cos \frac{\pi}{x} - 1 \right)$$

$$= \lim_{x \rightarrow \infty} \frac{x^2}{2} \left( -2 \sin^2 \frac{\pi}{2x} \right) = -\frac{\pi^2}{4}$$

5. **Ans. (B)**



Let height of cuboid be h and other side be 2a

$$\Rightarrow 2 - h^2 = 1 + a^2 \Rightarrow a^2 + h^2 = 1$$

Now, volume of cuboid =  $2 \cdot h \cdot 2a$

$$\Rightarrow v = 4ah$$

$$\therefore \frac{a^2 + h^2}{2} \geq \sqrt{a^2 \cdot h^2} \quad (\because \text{A.M.} > \text{G.M.})$$

$$\Rightarrow ah \leq \frac{1}{2}$$

maximum volume of cuboid = 2

$$(\text{when } a = \frac{1}{\sqrt{2}}, h = \frac{1}{\sqrt{2}})$$

6. **Ans. (A,B)**

**Sol.** 
$$I_n = \int_0^{\pi/2} x (\sin x + \cos x)^n dx = 2^{n/2} \int_0^{\pi/2} x \cos^n \left( x - \frac{\pi}{4} \right) dx$$

Put  $x - \frac{\pi}{4} = t$

$$\Rightarrow I_n = 2^{n/2} \int_{-\pi/4}^{\pi/4} \left( \frac{\pi}{4} + t \right) \cos^n t dt$$

$$\Rightarrow I_n = \pi 2^{\frac{n}{2}-1} \int_0^{\pi/4} (\cos t)^n dt$$

$$\Rightarrow I_n = \pi 2^{\frac{n-1}{2}} \left\{ \left( (\cos t)^{n-1} \sin t \right) \Big|_0^{\pi/4} + (n-1) \int_0^{\pi/4} (\cos t)^{n-2} (1 - \cos^2 t) dt \right\}$$

$$\Rightarrow \frac{n I_n - \pi/2}{I_{n-2}} = 2(n-1)$$

{After simplification}

7. **Ans. (A,B,D)**

**Sol.**  $f(x) = (x-1)|x-3| - 4x + 12$

$$f(x) = \begin{cases} 9-x^2 & x < 3 \\ x^2 - 8x + 15 & x \geq 3 \end{cases}$$

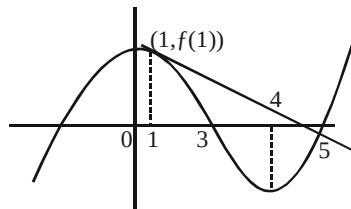
$$f'(x) = \begin{cases} -2x & x < 3 \\ 2x - 8, & x > 3 \end{cases}$$

$$f'(3^-) = -6$$

$$f'(3^+) = -2$$

$$f'(0) = 0$$

$$f'(4) = 0$$



8. **Ans. (A,B)**

**Sol.** We have  $hx + ky = (h^2 + k^2)$

$$\text{Now, } \left| \frac{ah - (h^2 + k^2)}{\sqrt{h^2 + k^2}} \right| \left| \frac{h^2 + k^2}{\sqrt{h^2 + k^2}} \right| = \lambda^2$$

$$\Rightarrow |ah - (h^2 + k^2)| = \lambda^2$$

$$\text{Since, } (ah - (h^2 + k^2))(h^2 + k^2) < 0$$

$$\Rightarrow ah - h^2 - k^2 < 0$$

$$\text{Hence, } h^2 + k^2 - ah = \lambda^2 \text{ So, } x^2 + y^2 - ax - \lambda^2 = 0$$

9. **Ans. (A,B)**

**Sol.**  $AB = B^T$

$$\Rightarrow A = B^T B^{-1}$$

$$|A| = |B^T| \cdot |B^{-1}| = 1 \quad \dots(1)$$

$$\text{Also } \text{Adj } A = \text{Adj}(B^T B^{-1})$$

$$= \text{Adj}(B^{-1}) \text{Adj}(B^T)$$

$$\text{Adj } A = (\text{Adj } B)^{-1} \text{Adj}(B^T)$$

$$\Rightarrow (\text{Adj } B)^T = (\text{Adj } B) \text{Adj } A$$

$$= (\text{Adj } B) B$$

$$\Rightarrow (\text{Adj } B)^T = |B| \cdot I$$

$$\Rightarrow \text{Adj } B = |B| \cdot I$$

$$\Rightarrow \text{Adj}(\text{Adj } A) = |\text{Adj } A| \cdot I$$

$$\therefore |A|^{n-2} A = |A|^{n-1} I \quad (\text{order } n)$$

$$\Rightarrow A = |A| I \Rightarrow A = I \quad (\text{By (1)})$$

$$\therefore B = I$$

10. **Ans. (A,B,C)**

**Sol.**  $A^T - I = A^T A$

take transpose on both the sides

$$\Rightarrow A - I = A^T A \Rightarrow A^T - I = A - I$$

$$\Rightarrow A^T = A \Rightarrow A \text{ is symmetric}$$

$$\text{Now } A - I = A^2 \Rightarrow A^2 - A + I = 0$$

Now, match the options.

11. **Ans. (A,C)**

**Sol.** Given equation

$$\sec^2(x+y)dx + 2\sin x dx$$

$$+ \sec^2(x+y)dy + 2\sec^2 y dy = 0$$

$$\sec^2(x+y)d(x+y) + 2\sin x dx + 2\sec^2 y dy = 0$$

$$\therefore \tan(x+y) - 2\cos x + 2\tan y + c = 0$$

$$\therefore y(0) = \frac{\pi}{4} \Rightarrow 1 - 2 + 2 + c = 0$$

$$\Rightarrow c = -1$$

$$\tan(x+y) - 2\cos x + 2\tan y - 1 = 0$$

$$\therefore y\left(\frac{\pi}{2}\right) = \frac{\pi}{4} \quad \left(\because y \in \left[0, \frac{\pi}{2}\right)\right)$$

Put  $x = \frac{\pi}{2}, y = \frac{\pi}{4}$  in differential equation

$$\Rightarrow \frac{dy}{dx} = -\frac{2}{3}$$

$\Rightarrow$  Equation of tangent to the curve at

$$\left(\frac{\pi}{2}, \frac{\pi}{4}\right) \text{ is } 8x + 12y = 7\pi$$

12. **Ans. (A,B,D)**

**Sol.**  $a + b + c = -1, ab + bc + ca = 0$

$$\Rightarrow a^2 + b^2 + c^2 = 1$$

Thus all three rows form three mutually perpendicular unit vector so A is orthogonal matrix.

13. **Ans. (A,B)**

**Sol.**  $V = |\vec{a} \cdot (\vec{b} \times \vec{c})| = |\vec{a}| |\vec{b} \times \vec{c}| \cos \alpha$

Similarly,

$$V = |\vec{b}| |\vec{c} \times \vec{a}| \cos \alpha \quad \& \quad V = |\vec{c}| |\vec{a} \times \vec{b}| \cos \alpha$$

$$\text{While } S = 2(|\vec{a} \times \vec{b}| + |\vec{b} \times \vec{c}| + |\vec{c} \times \vec{a}|)$$

$$= 2 \left( \frac{V}{|\vec{c}| \cos \alpha} + \frac{V}{|\vec{a}| \cos \alpha} + \frac{V}{|\vec{b}| \cos \alpha} \right)$$

$$\Rightarrow \frac{V}{S} = \frac{|\cos \alpha|}{2} \left[ \frac{1}{|\vec{a}|} + \frac{1}{|\vec{b}|} + \frac{1}{|\vec{c}|} \right]^{-1}$$

$$\text{Now, } V = |\vec{a} \cdot (\vec{b} \times \vec{c})| = |\vec{a}| |\vec{b} \times \vec{c}| \cos \alpha$$

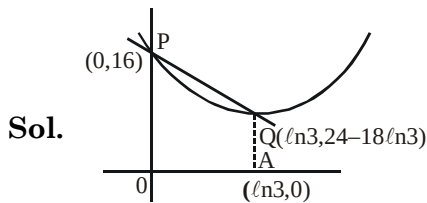
$$= |\vec{a}| |\vec{b}| |\vec{c}| \sin(\vec{b} \wedge \vec{c}) |\cos \alpha| \leq |\vec{a}| |\vec{b}| |\vec{c}|$$

Also,  $\sin(\vec{a} \wedge \vec{b}) = \sin(\vec{b} \wedge \vec{c}) = \sin(\vec{c} \wedge \vec{a})$

So, it not necessary that they are equally inclined.

### SECTION-II

1. Ans. 1.50



Area of trap OAQP =  $\frac{1}{2}(40 - 18 \ln 3) \ln 3$

Req. area

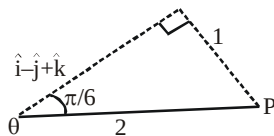
$$= \frac{1}{2}(40 - 18 \ln 3) \ln 3 - \int_0^{\ln 3} (e^{2x} - 18x + 15) dx$$

$$= 5 \ln 3 - 4$$

2. Ans. 3.14

Sol.  $(\hat{i} - \hat{j} + \hat{k}) \cdot (\hat{a}\hat{i} + \hat{b}\hat{j} + \hat{c}\hat{k}) = 3$

$$\cos \theta = \frac{\sqrt{3}}{2}$$



$$\sqrt{3} \cdot 2 \cos \theta = 3 \quad \{\because a^2 + b^2 + c^2 = 4\}$$

locus of 'P' is circle with radius  $A = \pi$   
 $[A] = 3$

3. Ans. 10.66 or 10.67

Sol. There will be 8 planes symmetrically located about origin.

Consider  $x + y + z = 2$

Volume enclosed with origin by this plane

$$= \frac{1}{6} \begin{vmatrix} 2\hat{i} & 2\hat{j} & 2\hat{k} \end{vmatrix} = \frac{8}{6} = \frac{4}{3}$$

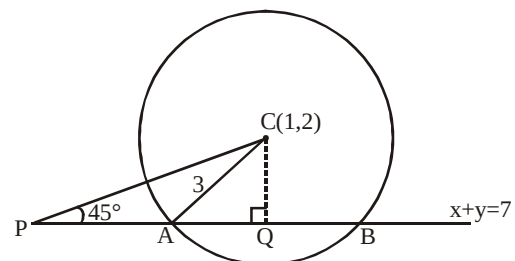
$$\therefore \text{Total volume} = \frac{8 \times 4}{3}$$

4. Ans. 5.64

Sol.  $CQ = \left| \frac{1+2-7}{\sqrt{2}} \right| = 2\sqrt{2}$

$\therefore$  triangle CPQ is right angle

$$\Rightarrow PQ = 2\sqrt{2}$$



Now, length of chord AB =  $2\sqrt{9-8} = 2$

$$\Rightarrow AQ = 1$$

5. Ans. 0.03

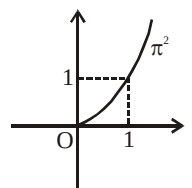
Sol.  $f(x) = \max\{x, x^2\} = \begin{cases} x & ; x \in [0, 1] \\ x^2 & ; x \in (1, \infty) \end{cases}$

$$\Rightarrow f \circ f(x) = \begin{cases} f(x) & ; 0 \leq f(x) \leq 1 \\ f^2(x) & ; f(x) > 1 \end{cases}$$

$$= \begin{cases} x & 0 \leq x \leq 1 \\ x^4 & x > 1 \end{cases}$$

$$(f \circ f)^{-1}(x) = x^{1/4} \text{ for } x > 1$$

$$\frac{d}{dx} (f \circ f)^{-1}(x) = \frac{1}{4} x^{-3/4} \text{ put } x = 16$$



**JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)****PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I****1. Ans. (B)****Sol.** By faraday's Law

$$\varepsilon = -\frac{d\phi}{dt} = -\left(d \times \frac{d}{2}\right) \frac{d}{dt}(Kt) = -\frac{Kd^2}{2}$$

 $\varepsilon$  is emf induced in each half.

$$\therefore Q_1 = -C\varepsilon \text{ and } Q_2 = 2C\varepsilon$$

After central wire is cut and magnetic field is kept constant, there is no flux change. Hence total charge distributed on both capacitor until potential difference across each capacitor becomes same.

$$\therefore \text{Total charge } Q = Q_1 + Q_2 = C\varepsilon = \frac{Kd^2}{2}C$$

$$\therefore v = \frac{Q}{C_{eq}} = \frac{\frac{Kd^2}{2}C}{2C + C} = \frac{Kd^2}{6}$$

$$\therefore \text{charge on } C = Cv = \frac{KCd^2}{6}$$

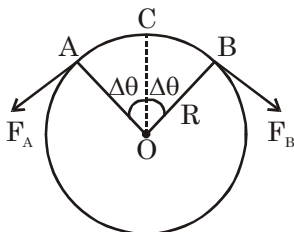
**2. Ans. (C)****Sol.** To find tension in the rope, net force along CO,

$$= F_A \sin \Delta\theta + F_B \sin \Delta\theta$$

(Components perpendicular to CO cancel each other)

But  $F_A = F_B = F$ , tension in the string.

$$\therefore 2F \sin \Delta\theta = m \times AB \times R\omega^2,$$

AB =  $R2\Delta\theta$  and  $m$  = linear density of the rope.

$$= 2mR^2\omega^2\Delta\theta$$

$$\text{For } \Delta\theta \ll 1, F = mR^2\omega^2$$

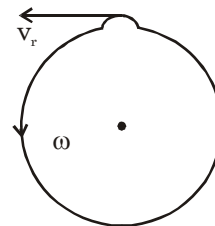
Now a kink is formed, moving in anticlockwise direction relative to an

observer in the ground speed of the kink

$$\text{relative to the rope } v_k = \sqrt{\frac{F}{m}} = R\omega$$

For an observer on the ground,

$$v_g = v_k + v_r = 60 \text{ ms}^{-1} (\text{given})$$



$$\therefore 2\omega R = 60 \text{ or } \omega R = 30 \text{ ms}^{-1}$$

$$\text{Hence tension } F = 0.2 \times 30 \times 30 = 180 \text{ N}$$

**3. Ans. (D)**

$$\text{Sol. } I_0 = \frac{V}{R} = \frac{12}{50} = .24 \text{ A} = 240 \text{ mA}$$

Current at any time 't'

$$I = I_0 e^{-t/\tau}, \tau = \frac{L}{R}$$

$$\Rightarrow 10 = 240 e^{-t/\tau}$$

$$\Rightarrow t = \tau \ln 24$$

from given question,

$$t = 5 \text{ s}$$

$$\therefore 5 = \frac{2}{R} \ln 24$$

$$R = \frac{2}{5} \ln 24 = \frac{2}{5} \times 3.178 = 1.271 \Omega$$

**4. Ans. (C)**

$$\text{Sol. } y = y_1 + y_2$$

$$\Rightarrow y = 2a \sin \left( \frac{\pi}{2}x - \omega t \right) +$$

$$a \sin \left( \frac{\pi}{2}x + \omega t + \frac{\pi}{3} \right)$$

$$\Rightarrow y = a \sin \left( \frac{\pi}{2}x + \frac{\pi}{6} \right) \cdot \cos \left( \omega t + \frac{\pi}{6} \right)$$

Nodes are positions where,

$$A = 2a \sin \left( \frac{\pi}{2}x + \frac{\pi}{6} \right) = 0$$

$$\Rightarrow \frac{\pi}{2}x + \frac{\pi}{6} = \pi, 2\pi, 3\pi, \dots$$

$$\therefore \frac{x}{2} = \frac{5}{6}, \frac{11}{6}, \frac{17}{6}, \dots$$

$$\Rightarrow x = \frac{5}{3}, \frac{11}{3}, \frac{17}{3}, \dots$$

$$\therefore 0 \leq x \leq 6, \therefore \text{no. of nodes} = 3$$

5. **Ans. (C)**

**Sol.**  $8 = \frac{100}{100\pi L} \Rightarrow L = \frac{1}{8\pi} \text{ Henry}$

and  $R = 100/10 \Rightarrow R = 10\Omega$

when connected in parallel, let  $I_1$  and  $I_2$  be currents then.

$$I = \sqrt{I_1^2 + I_2^2}$$

$$I_1 = \frac{E_v}{\omega L} = \frac{150}{\frac{1}{8\pi} \times 2\pi \times 40} = 15 \text{ Amp.},$$

$$I_2 = \frac{150}{10} = 15 \text{ Amp}$$

$$I = 15\sqrt{2} \text{ Amp.}$$

6. **Ans. (C)**

**Sol.** Series combination of  $R$  and  $L$  is,

$$Y = \frac{1}{10 + i10} = \frac{10 - i10}{200} = \frac{1}{20} - \frac{i}{20}$$

It is equivalent to parallel combination of resistance ( $R = 20\Omega$ ) and inductor ( $X_L = 20\Omega$ ). so combination to be resistive,  $x_L = x_C$

$$\Rightarrow 20 = \frac{1}{2 \times 10^3 C} \Rightarrow C = 25\mu\text{F}$$

7. **Ans. (A, C)**

**Sol.** When current through  $L$  is maximum

$$\frac{dI}{dt} = 0$$

$\Rightarrow$  emf induced in  $L$  is zero at that instant.

At this time a total charge  $Q = 9CV_0 - 2CV_0$

$\Rightarrow 7CV_0$  is distributed on the two capacitors so that potential difference across them is same ( $V$ )

$$CV + 3CV = 7CV_0$$

$$\Rightarrow V = \frac{7V_0}{4}$$

Energy conservation

$$\frac{1}{2}LI_{\max}^2 = \left[ \frac{1}{2}C(2V_0)^2 + \frac{1}{2}3C(3V_0)^2 \right] - \left[ \frac{1}{2}CV^2 + \frac{1}{2}(3C)V^2 \right]$$

$$\text{Solving } I_{\max} = \frac{5V_0}{2} \sqrt{\frac{3C}{L}}$$

8. **Ans. (B, D)**

**Sol.** Superconducting loops have zero resistance, therefore when loops are brought together, the magnetic flux which passes through each of them, must not change otherwise there would be a voltage across each loop, which would cause an infinite current to avoid infinite current the current in the loop changes so as to keep flux fixed. When loops are far apart the flux passing through loop,

$$\phi_1 = LI \quad \dots(1)$$

when the two loops are brought close together, the mutual inductance is equal to the self inductance

$$\therefore \phi_f = LI_f + LI_f = 2LI_f$$

$$\therefore \phi_i = \phi_f$$

$$\therefore I_f = \frac{I}{2}$$

9. **Ans. (A, B, D)**

**Sol.** Loop equation gives

$$L \frac{di}{dt} + iR = \frac{\mu_0 Na^2 \omega}{2\ell} i$$

$$\text{So } \frac{di}{dt} = \frac{1}{L} \left( \frac{\mu_0 Na^2 \omega}{2\ell} - R \right) i$$

$$\text{So to grow current always, } \frac{\mu_0 Na^2 \omega}{2\ell} - R > 0$$

$$\Rightarrow \omega > \frac{2\ell R}{\mu_0 Na^2}$$

10. **Ans. (A, B, D)**

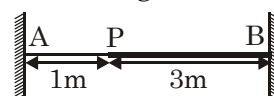
**Sol.** Mass per unit length of light string

$$\mu = \frac{10 \times 10^{-3}}{1} = 10^{-2} \text{ kgm}^{-1}$$

Mass per unit length of heavy string

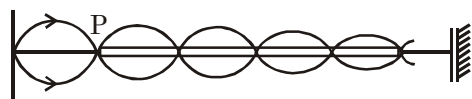
$$= \frac{270 \times 10^{-3}}{3} = 9 \times 10^{-2} \text{ kgm}^{-1} = 9\mu$$

Tension  $T = 16 \text{ N}$ .



(i) Let  $n_1$  and  $n_2$  be the number of loops,  $\lambda_1$  and  $\lambda_2$  be the wavelengths and  $v_1$  and  $v_2$  be the speeds in light and heavy strings respectively and  $f$  be the frequency of vibration. Then, for node P,

$$n_1 \frac{\lambda_1}{2} = 1m, \quad n_2 \frac{\lambda_2}{2} = 3m, \quad n_1, \quad n_2 \text{ being integers}$$



$$\Rightarrow n_2 \lambda_1 = 3n_1 \lambda_2$$

11. Ans. (A, C)

Sol. The magnetic force does not cause any change in speed and the frictional force will always be against the velocity.

$$ma = -kv$$

$$v \frac{dv}{dx} = -\frac{k}{m} v$$

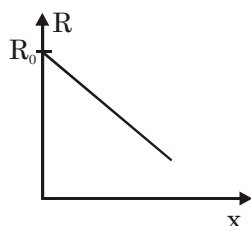
$$\int_{v_0}^v dv = -\frac{k}{m} \int_0^x dx$$

$$v - v_0 = -\frac{kx}{m}$$

$$v = v_0 - \frac{kx}{m}$$

$$\Rightarrow \therefore R = \frac{m\ell}{qB} = \frac{m}{qB_0} \left( v_0 - \frac{kx_0}{m} \right)$$

$$= R_0 - \frac{k}{qB_0} x$$

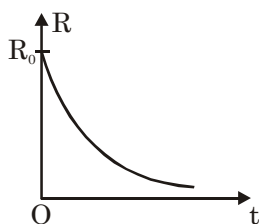


$$\text{Also } m \frac{dv}{dt} = -kv$$

$$\int_{v_0}^v \frac{dv}{v} = -\frac{k}{m} \int_0^t dt$$

$$\ln \frac{v}{v_0} = -\frac{kt}{m}$$

$$v = v_0 e^{-\frac{kt}{m}}$$



$$R = R_0 e^{-\frac{kt}{m}} \text{ where } R_0 = \frac{mv_0}{qB} = \text{initial radius}$$

12. Ans. (B, C)

Sol.  $d\vec{B}_{12}$  = Magnetic field at any point on the elementary element AB due to current carrying element  $I_2 d\vec{\ell}_2$ .

$$d\vec{B}_{12} = \frac{\mu_0}{4\pi} \left( \frac{I_2 d\vec{\ell}_2 \times \vec{r}_{12}}{r_{12}^3} \right)$$

$$d\vec{F}_{12} = I_1 d\vec{\ell}_1 \times d\vec{B}_{12} = \left( \frac{\mu_0}{4\pi} \right) \frac{I_1 d\vec{\ell}_1 \times (I_2 d\vec{\ell}_2 \times \vec{r}_{12})}{r_{12}^3}$$

The force  $d\vec{F}_{12}$  will be in the plane of loop and perpendicular to the current carrying element  $I_1 d\vec{\ell}_1$  (say along  $\vec{p}_{12}$ ).

$$\text{Similarly } d\vec{F}_{21} = \frac{\mu_0}{4\pi} \frac{I_2 d\vec{\ell}_2 \times (I_1 d\vec{\ell}_1 \times \vec{r}_{21})}{r_{21}^3}$$

[Say along  $\vec{p}_{21}$ ]

$\therefore \hat{p}_{12}$  is not parallel to  $\hat{p}_{21}$ , in general

So,  $d\vec{F}_{12} + d\vec{F}_{21} \neq \vec{0}$

$$\text{But } \vec{F}_{12} = \frac{\mu_0 I_1 I_2}{4\pi} \int_{L_1} \int_{L_2} \frac{d\vec{\ell}_1 \times (d\vec{\ell}_2 \times \vec{r}_{12})}{r_{12}^3}$$

$$\vec{F}_{21} = \frac{\mu_0 I_1 I_2}{4\pi} \int_{L_2} \int_{L_1} \frac{d\vec{\ell}_2 \times (d\vec{\ell}_1 \times \vec{r}_{21})}{r_{21}^3}$$

Hence,  $\vec{F}_{12} + \vec{F}_{21} = \vec{0}$

13. Ans. (A, C)

Sol. Let equation of the wave be

$$y = A \sin kx \cos \omega t$$

$$\therefore 5\sqrt{3} = A \sin(k \times 10) \dots (1)$$

$$\text{and } 5\sqrt{3} = A \sin(k \times 20) \dots (2)$$

From (1) and (2), and the given conditions,

$$20k = \pi - 10k \Rightarrow k = \frac{\pi}{30}$$

$$(a) 5\sqrt{3} = A \sin \left( \frac{\pi}{30} \times 10 \right)$$

using equation (1)

$$\Rightarrow A = 10 \text{ mm}$$

$$(b) \therefore k = \frac{2\pi}{\lambda} = \frac{\pi}{30},$$

$$\Rightarrow \lambda = 60 \text{ cm},$$

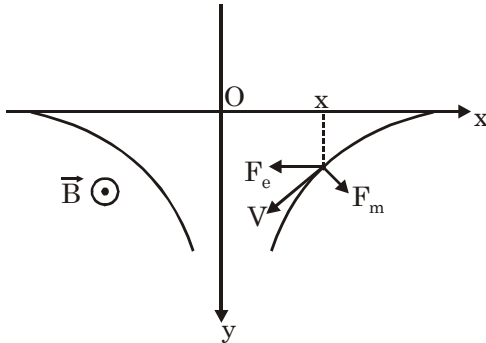
$$\therefore L = n \frac{\lambda}{2}$$

$$\Rightarrow n = \frac{2L}{\lambda} = \frac{2 \times 150}{60} = 5$$

As the number of loops is 5 the string vibrates in the 4<sup>th</sup> overtone.

14. Ans. (A, D)

Sol. The two particles follow symmetrical curved paths. As the electrostatic attraction speeds them, the magnetic force curves the path. Magnetic force does not perform any work, therefore conservation of energy gives.



$$\frac{1}{2}mV^2 \times 2 - k \frac{q^2}{(2x)} = -k \frac{q^2}{(2a)}$$

$$\Rightarrow 2mV^2 = kq^2 \left( -\frac{1}{a} + \frac{1}{x} \right)$$

$$\Rightarrow V = \sqrt{\frac{kq^2}{2m} \left( \frac{1}{x} - \frac{1}{a} \right)}$$

Force on  $-q$  is

$$\vec{F} = -F_e \hat{i} + (-q)(V_x \hat{i} + V_y \hat{j}) \times (-B \hat{k})$$

$$= (-F_e + qV_x B) \hat{i} - qBV_y \hat{j}$$

$$\therefore m \frac{dV_y}{dt} = -qBV_x \quad m \frac{dV_x}{dt} = -qB \frac{dx}{dt}$$

$$\Rightarrow \int_0^{V_x} dV_x = -\frac{qB}{m} \int_a^x dx \Rightarrow V_x = \frac{qB}{m}(a-x)$$

15. Ans. (B)

Sol. For just flip over  $\tau_B \geq \tau_g$

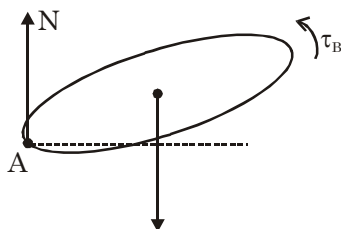
$$I \times \pi r^2 \times B \geq mgr$$

$$I_{\min} = I_0 = \frac{mgr}{\pi r^2 B} \Rightarrow m = \frac{\pi I_0 r B}{g}$$

16. Ans. (C)

Sol. Just after magnetic field I is switched on

$$\tau_A = I_A \alpha$$



$$\Rightarrow \pi r^2 \times I \times B - mgr = \frac{3}{2}mr^2\alpha$$

$$\Rightarrow \pi r^2 \times 2I_0 B - \pi I_0 r^2 B = \frac{3}{2}mr^2\alpha$$

$$\Rightarrow \pi r^2 I_0 B = \frac{3}{2}mr^2\alpha$$

$$\Rightarrow \alpha = \frac{2}{3} \frac{\pi}{m}$$

$$\text{Also } N - mg = ma_{\text{cm}}$$

$$N = mg + mr\alpha$$

$$= \pi I_0 r B + \frac{2}{3} \pi I_0 r B = \frac{5}{3} \pi I_0 r B$$

17. Ans. (A)

18. Ans. (B)

Sol. When current is 5A, the impedance is

$$Z = \frac{V}{I} = \frac{100}{5} = 20\Omega$$

This is equal to R.

Hence for  $\omega_0 = 1000$  rad/s

$$Z = R = 20\Omega$$

This is possible only if  $S_1$  is closed and

$$\omega_0 L = \frac{1}{\omega_0 C}$$

$$\Rightarrow LC = \frac{1}{10^6} \quad \dots(i)$$

When  $\omega = 500$  rad/s

$$I = 4A \Rightarrow \frac{100}{Z} = 4A$$

$$\Rightarrow Z = 25\Omega$$

$$\sqrt{R^2 + X^2} = 25 \Rightarrow X = 15\Omega$$

$$\therefore \frac{1}{\omega C} - \omega L = 15$$

[Note that X is not  $\omega L - \frac{1}{\omega C}$ , since on increasing the frequency from 500 rad/s to 1000 rad/s the impedance is decreasing]

$$\Rightarrow \frac{1}{500C} - 500L = 15 \quad \dots(ii)$$

Solving (i) and (ii)  $L = 10$  mH,  $C = 100$   $\mu$ F

**PART-2 : CHEMISTRY**

**SOLUTION**

**SECTION-I**

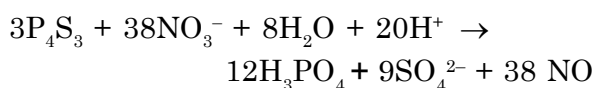
1. **Ans.(C)**

2. **Ans.(B)**

3. **Ans.(B)**

4. **Ans. (B)**

5. **Ans.(B)**



$$\frac{r_{\text{P}_4\text{S}_3}}{3} = \frac{r_{\text{H}_2\text{O}}}{8}$$

6. **Ans.(A)**

$$\text{Mass of solution} = 0.4 \times 32 + 0.6 \times 18 = 23.6 \text{ gm}$$

$$\text{Vol. of solution} = \frac{23.6}{0.9} = 26.22 \text{ cm}^3$$

$$V_{\text{soln.}} = X_1 V_1 + X_2 V_2$$

$$26.22 = 0.4 \times 40 + 0.6 \times V_2$$

$$V_2 = 17.03 \text{ cm}^3 / \text{mol}$$

7. **Ans.(A,B,C)**

8. **Ans.(A,D)**

9. **Ans.(A,B,D)**

$$\frac{0.496}{M_w} \times 1 = \frac{1}{2} \times 2 \times \frac{40}{1000}$$

$$M_w \text{ of base} = 12.4$$

For I solution

$$\Delta T_f = i K_f m$$

$$0.165 = i \times 1.86 \times \frac{1.5}{12.4 \times 1.5}$$

$$i = 1.1 = 1 + \alpha \Rightarrow \alpha = 0.1$$

$$\pi = i CRT = 2.13 \text{ atm}$$

10. **Ans.(A,C,D)**

11. **Ans.(B,C)**

12. **Ans.(A,B,C)**

13. **Ans.(A,B)**

14. **Ans.(A)**

15. **Ans.(B)**

16. **Ans.(C)**

17. **Ans.(B)**

18. **Ans.(A)**

**PART-3 : MATHEMATICS**

**SOLUTION**

**SECTION-I**

1. **Ans. (A)**

**Sol.** 
$$f'(x) = \frac{\left(1 + e^{\frac{1}{x}}\right) \cdot 1 + \frac{e^{1/x}}{x}}{\left(1 + e^{\frac{1}{x}}\right)^2}$$

$$\Rightarrow x \left(1 + e^{1/x}\right)^2 - f'(x) = x \left(1 + e^{1/x}\right) + e^{1/x}$$

$$\Rightarrow x \cdot \frac{x^2}{f^2(x)} \cdot f'(x) = x \cdot \frac{x}{f(x)} + e^{1/x}$$

$$\Rightarrow x \cdot \frac{x^2}{f^2(x)} \cdot f'(x) = x \cdot \frac{x}{f(x)} + e^{1/x}$$

$$\Rightarrow x^3 f'(x) = x^2 f(x) + e^{1/x} f^2(x)$$

2. **Ans. (D)**

**Sol.** Equation of plane  $\frac{x}{\alpha'} + \frac{y}{\beta'} + \frac{z}{\gamma'} = 1$

$$\text{equation of diagonal : } \frac{x}{\alpha} = \frac{y}{\beta} = \frac{z}{\gamma}$$

$$Q' \equiv \frac{\lambda\alpha}{\alpha'} + \frac{\lambda\beta}{\beta'} + \frac{\lambda\gamma}{\gamma'} = 1$$

$$\lambda = \frac{1}{\frac{\alpha}{\alpha'} + \frac{\beta}{\beta'} + \frac{\gamma}{\gamma'}}$$

$$\frac{PQ}{PQ'} = \frac{\sqrt{\alpha^2 + \beta^2 + \gamma^2}}{\lambda \sqrt{\alpha^2 + \beta^2 + \gamma^2}} = \frac{\alpha}{\alpha'} + \frac{\beta}{\beta'} + \frac{\gamma}{\gamma'}$$

3. **Ans. (D)**

**Sol.** Given that  $\vec{p} \times \vec{q} = \vec{r}$

$$\Rightarrow (\vec{p} \times \vec{q}) \cdot \vec{r} = |\vec{r}|^2$$

$$\Rightarrow [\vec{p} \vec{q} \vec{r}] = 1$$

$$\text{Let } \vec{a} = x\vec{p} + y\vec{q} + z\vec{r} \quad \dots\dots(1)$$

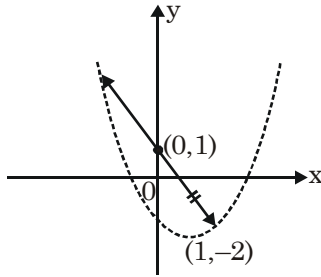
Take dot product in equation (1) both sides

with  $\vec{p} \times \vec{q}$ ,  $\vec{q} \times \vec{r}$  and  $\vec{r} \times \vec{p}$  we get

$$x = 1, y = 2, z = 3.$$

4. Ans. (B)

Sol.  $(y + 2) = (x - 1)^2$  and  $(y - 1) = -mx$  which passes through  $(0, 1)$



for minimum area  $(0, 1)$  must be the mid point of chord.

5. Ans. (D)

Sol.  $I = \int \frac{\operatorname{cosec}^2 x}{(\operatorname{cosec} x + \cot x)^{9/2}} dx$

Put  $\operatorname{cosec} x + \cot x = z$

$\operatorname{cosec} x - \cot x = \frac{1}{z}$

$-2\operatorname{cosec}^2 x dx = \left(1 + \frac{1}{z^2}\right) dz$

$\therefore I = -\frac{1}{2} \int \frac{1 + \frac{1}{z^2}}{z^{9/2}} dz = -\frac{1}{2} \left[ \int z^{-9/2} dz + \int z^{-11/2} dz \right]$

$= -\frac{1}{2} \left[ \frac{z^{-7/2}}{(-7/2)} + \frac{z^{-11/2}}{(-11/2)} \right] + C$

$= z^{-7/2} \left[ \frac{1}{7} + \frac{z^{-3}}{11} \right] + C$

$= (\operatorname{cosec} x - \cot x)^{7/2} \left( \frac{1}{7} + \frac{(\operatorname{cosec} x - \cot x)^2}{11} \right) + C$

6. Ans. (A)

Sol. Consider  $(M^2 + 2M + 2I)(M^2 - 2M + 2I)$

$\therefore (M^2 + 2M + 2I)(M^2 - 2M + 2I) = 4I$

$\therefore$  Neither of determinant value will be zero.

7. Ans. (A,B,C,D)

Sol. Given

$f(x) = e^x \int_0^x e^{-y} f'(y) dy - (x^2 - x + 1)e^x \dots (1)$

$\Rightarrow f'(x) = e^x \cdot e^{-x} \cdot f'(x) + e^x \int_0^x e^{-y} f'(y) dy - (x^2 - x + 1)e^x - e^x(2x - 1)$

or,  $e^x \int_0^x e^{-y} f'(y) dy = (x^2 - x + 1)e^x + e^x(2x - 1)$

Now using (1)

$f(x) + (x^2 - x + 1)e^x = (x^2 - x + 1)e^x + e^x(2x - 1)$

$\Rightarrow f(x) = e^x(2x - 1)$ , Now verify alternatives.

8. Ans. (A,B,C,D)

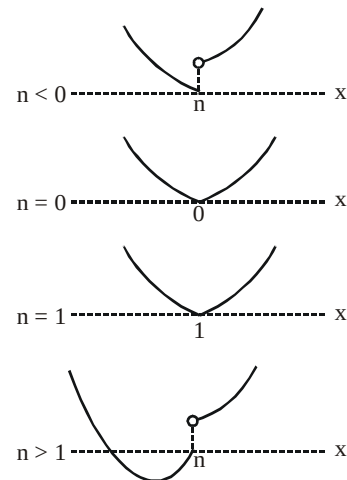
Sol.  $|\vec{v}_1 + \vec{v}_2 - \vec{v}_3|^2 + |\vec{v}_2 + \vec{v}_3 - \vec{v}_1|^2 + |\vec{v}_3 + \vec{v}_1 - \vec{v}_2|^2 = 36$

$\Rightarrow \vec{v}_1 + \vec{v}_2 + \vec{v}_3 = \vec{0}$

So,  $\vec{v}_1, \vec{v}_2, \vec{v}_3$  are coplanar vectors.

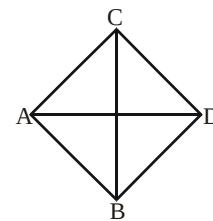
9. Ans. (A,D)

Sol.



10. Ans. (A,C)

Sol. Let centroid is M



$\vec{AB} = \vec{MB} - \vec{MA} = \hat{i} + \hat{j} - 6\hat{k}$

$\vec{AD} = \vec{MD} - \vec{MA} = -\hat{j} - 4\hat{k}$

$\vec{AC} = \vec{MC} - \vec{MA} = -5\hat{i} - 8\hat{j} + 2\hat{k}$

(A) Volume  $= \frac{1}{6} [\vec{AB} \cdot \vec{AD} \times \vec{AC}] = \frac{8}{3}$

(B)  $AB = \sqrt{38}$ ,  $AD = \sqrt{17}$

(C) Area  $= \frac{1}{2} |\vec{AB} \times \vec{AD}| = \frac{\sqrt{117}}{2}$

11. Ans. (A,B,C)

Sol. (A)  $\int_0^1 f(x) dx + \int_0^e g(x) dx = e - 0$

$\Rightarrow \int_0^e g(x) dx = e - I$



**Paragraph for Question 17 & 18**

**Sol.**  $2y \operatorname{cosec} 2x \cdot dx + \ln \cot y dx + \ln \tan x \cdot dy - 2x \operatorname{cosec} 2y dy = 0$

$$\Rightarrow (2y \operatorname{cosec} 2x dx + \ln \tan x dy) + (-2x \operatorname{cosec} 2y dy + \ln \cot y dx) = 0$$

$$\Rightarrow d(y \ln \tan x) + d(x \ln \cot y) = 0$$

$$\Rightarrow y \ln \tan x + x \ln \cot y = c$$

$$\Rightarrow (\tan x)^y \cdot (\cot y)^x = k$$

$$f\left(\frac{\pi}{4}, \frac{\pi}{2}\right) = 0 \Rightarrow k = 0$$

$$f(x, y) = (\tan x)^y (\cot y)^x$$

**17. Ans. (B)**

**18. Ans. (C)**

**Sol.**  $\int_0^{\infty} \frac{\cos x}{x} dx = \frac{\pi}{2}$

$$\int_0^{\infty} \frac{\cos^3 x}{x} dx = \int_0^{\infty} \frac{3 \cos x}{4} dx + \int_0^{\infty} \frac{\cos 3x}{4x} dx$$

$$\Rightarrow \frac{3}{4} \cdot \frac{\pi}{2} + \int_0^{\infty} \frac{\cos t}{4 \cdot \frac{t}{3}} \frac{dt}{3}$$

$$\Rightarrow \frac{3}{4} \cdot \frac{\pi}{2} + \frac{1}{4} \cdot \frac{\pi}{2} = \frac{\pi}{2}$$